

# **Week 3 Project Report**

**Project Title: Shopify Stock Performance Dashboard**

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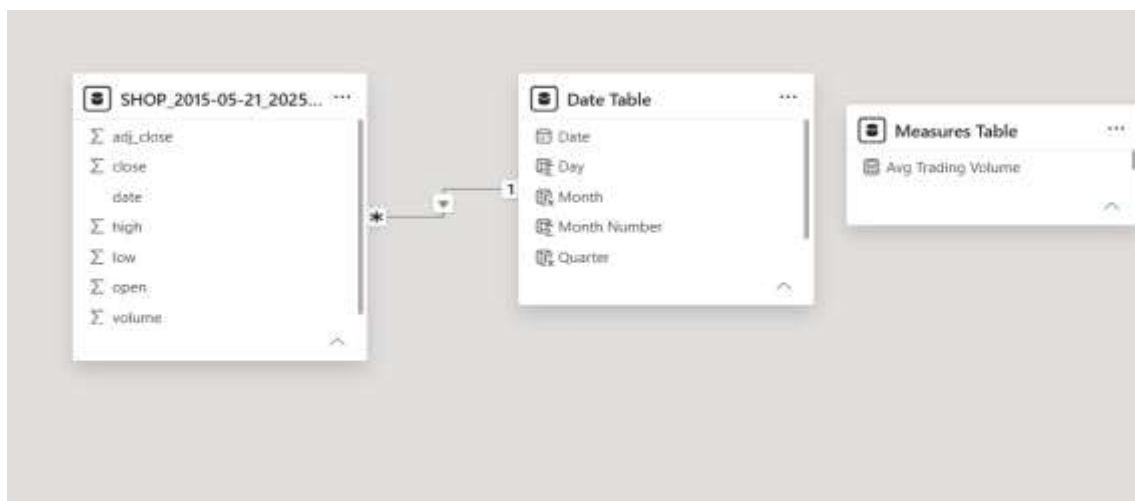
## ◆ Introduction

In this project, I built a dashboard in Power BI to study **Shopify stock performance**. The goal was to show the latest values, daily changes, yearly growth, and trading activity in a clear and interactive way.

## ◆ Step 1: Data Preparation

- Imported the **Shopify stock dataset** (Date, Open, Close, High, Low, Volume).
- Created a **Date Table** using CALENDAR().
- Marked the Date Table as the official date table.
- Linked **Date Table[Date]** with **SHOP[date]**.

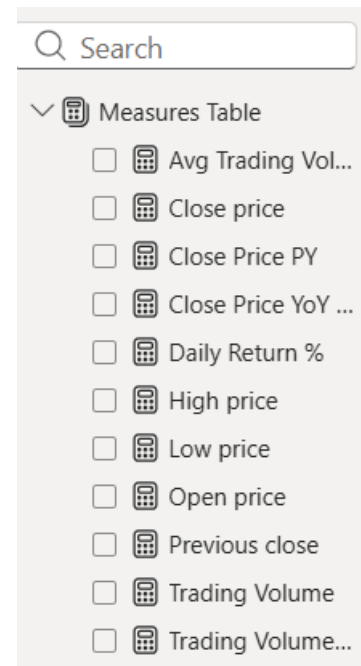
👉 This setup allowed me to use time intelligence functions like daily return and year-over-year comparison.

[illegible]

## ◆ Step 2: Measures

I wrote DAX formulas to calculate important values:

- Avg Trading Volume = `AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume])`
- Close price = `AVERAGE('SHOP_2015-05-21_2025-03-16'[close])`
- Close Price PY = `CALCULATE([Close price],SAMEPERIODLASTYEAR('Date Table'[Date]))`
- Close Price YoY % = `DIVIDE([Close price]-[Close Price PY],[Close Price PY],0)`
- Daily Return % =
- `VAR LatestDate = LASTDATE('Date Table'[Date])`
- `VAR CurrentClose =`
- `CALCULATE([Close Price], 'Date Table'[Date] = LatestDate)`
- `VAR PrevClose =`
- `CALCULATE([Close Price], DATEADD('Date Table'[Date], -1, DAY))`
- `RETURN`
- `DIVIDE(CurrentClose - PrevClose, PrevClose)`
- 
- High price = `MAX('SHOP_2015-05-21_2025-03-16'[High])`
- Low price = `MIN('SHOP_2015-05-21_2025-03-16'[low])`
- Open price = `AVERAGE('SHOP_2015-05-21_2025-03-16'[Open])`



- Previous close = `CALCULATE([Close price],DATEADD('Date Table'[Date], -1, DAY))`
- Trading Volume = `SUM('SHOP_2015-05-21_2025-03-16'[volume])`
- Trading Volume YTD = `TOTALYTD([Trading Volume], 'Date Table'[Date])`

👉 These measures gave me the numbers for my visuals.

### ◆ Step 3: KPI Cards (Top Row)

I placed **four cards** at the top row:

1. Close Price
2. Daily Return % (latest)
3. Close Price YoY %
4. Avg Trading Volume

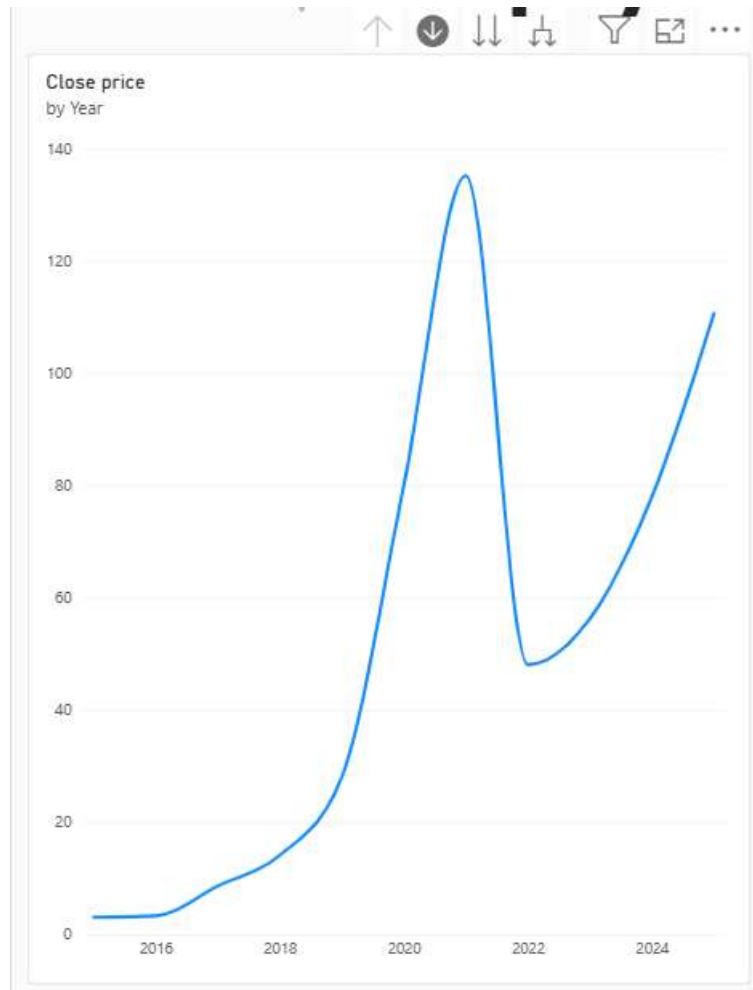
👉 These cards give a quick snapshot of the stock's performance.



### ◆ Step 4: Line Chart (Middle)

- Added a **Line Chart** with Date hierarchy on the X-axis and Close Price on the Y-axis.
- Enabled **drilldown and drillup** (Year → Quarter → Month → Day).

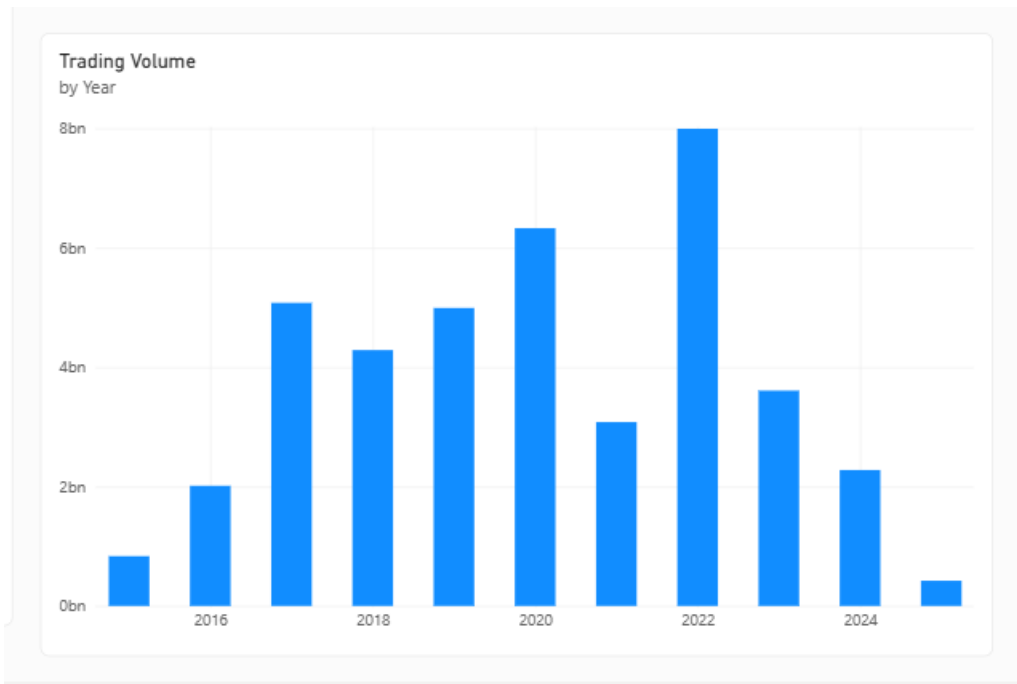
👉 This chart shows how the stock price changes over time and lets me zoom into details.



#### ◆ Step 5: Trading Volume Chart (Bottom)

- Added a **Clustered Column Chart** with Date on the X-axis and Trading Volume on the Y-axis.
- Enabled **drilldown and drillup** here too, so I can see yearly, monthly, or daily trading activity.

👉 This chart shows how much trading happened and supports the price trend analysis.



#### ◆ Step 6: Slicer (Right Side)

- Inserted a **Date slicer** on the right side.
- Title → *Select Date Range*.
- Format → Between slider.

👉 This slicer filters the entire dashboard based on the chosen time period.

Date, Year, Month, Day

- ✓ ☐ 21-05-2015 ...
- ✓ ☐ 22-05-2015 ...
- ✓ ☐ 23-05-2015 ...
- ✓ ☐ 24-05-2015 ...
- ✓ ☐ 25-05-2015 ...
- ✓ ☐ 26-05-2015 ...
- ✓ ☐ 27-05-2015 ...
- ✓ ☐ 28-05-2015 ...
- ✓ ☐ 29-05-2015 ...
- ✓ ☐ 30-05-2015 ...
- ✓ ☐ 31-05-2015 ...
- ✓ ☐ 01-06-2015 ...
- ✓ ☐ 02-06-2015 ...
- ✓ ☐ 03-06-2015 ...
- ✓ ☐ 04-06-2015 ...
- ✓ ☐ 05-06-2015 ...

